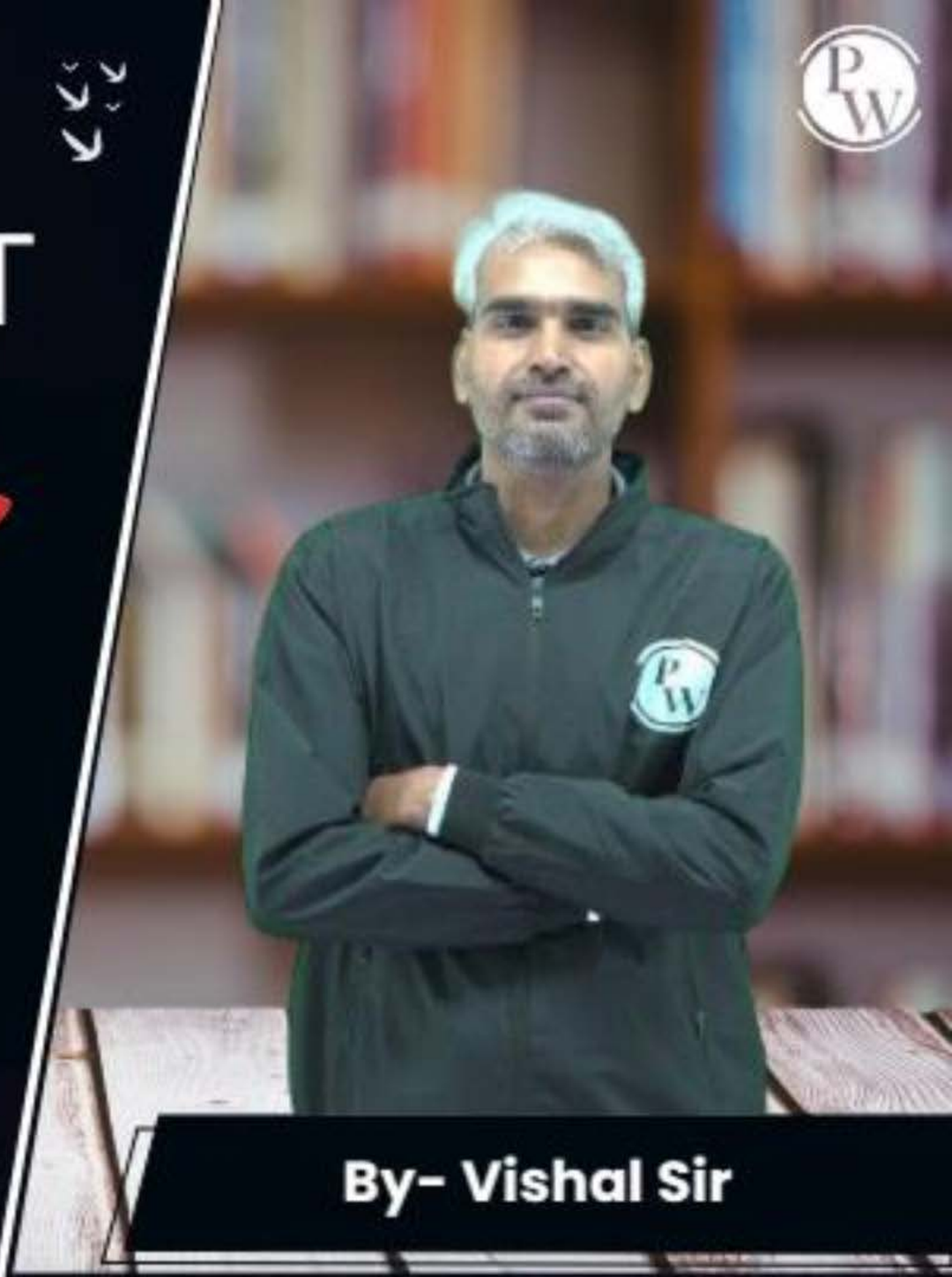
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 07



By- Vishal Sir

Recap of Previous Lecture



Topic

Planar graph



Slide

Topics to be Covered



- Topic: Euler's equation for disconnected planar graph
- Topic: Vertex coloring
- Topic: Chromatic number
- Topic: Edge coloring

Slide

Today's Topic

H.W. :- Obtain Euler's Equation for disconnected Planar graph with 'k' Connected Components in it.

* Euler's equation for a disconnected planar graph with 'k' Connected Components

Euler's equation for disconnected Planar graph

{may or may not be simple}

- Let 'G' is a planar graph with 'K' Connected Components.
- Let |V| = Total no. of vertices in G.
  |R| = Total no. of regions in G = (No. of all interior regions + 1)
  |E| = Total no. of edges in G
  K : No. of Connected Components in G.

{∵ Each Connected Component will also be planar} graph is planar

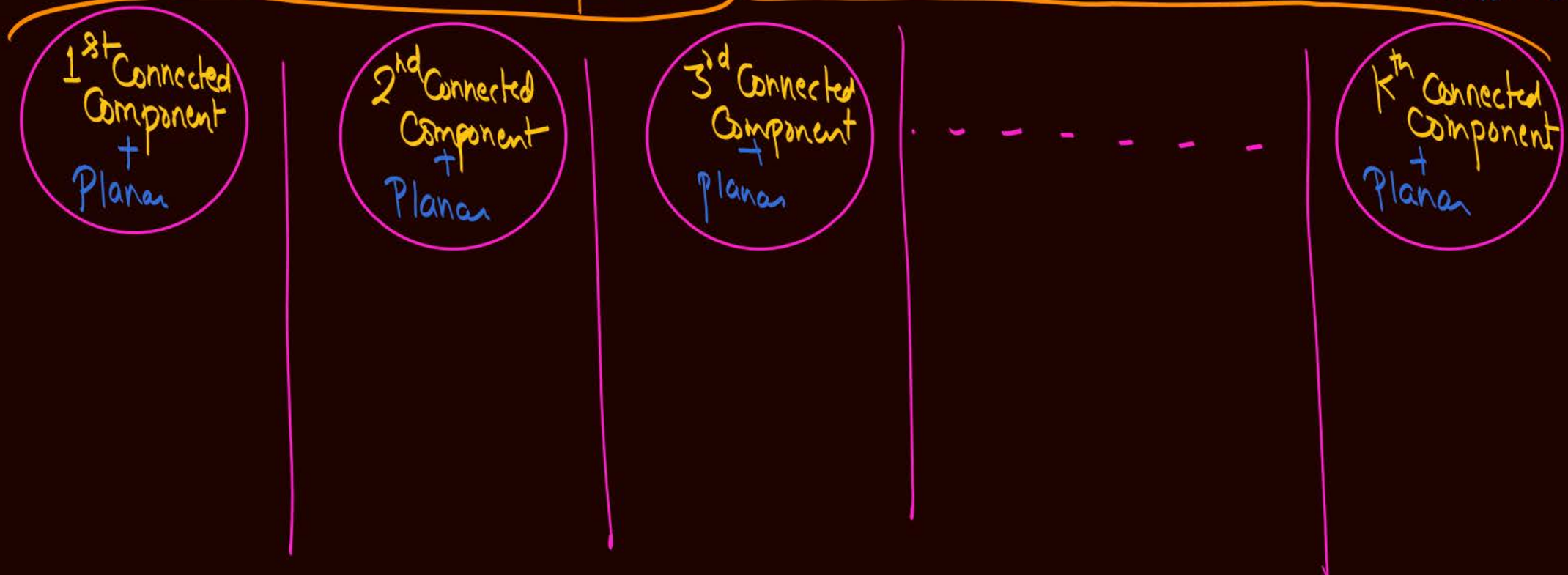
Euler's equation for disconnected Planar graph

{may or may not be simple}

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Graph 'G' with 'K' Connected Components



Euler's equation for disconnected Planar graph

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Graph 'G' with 'K' Connected Components

{∴ Each Connected Component will also be planar} graph is planar

1st Connected Component + Planar

Let $n = |V_1|$
$e = |E_1|$
$f = |R_1|$
∴ $|V_1| + |R_1| = |E_1| + 2$

2nd Connected Component + Planar

Let $n = |V_2|$
$e = |E_2|$
$f = |R_2|$
$|V_2| + |R_2| = |E_2| + 2$

3rd Connected Component + planar

. . . . . .

Kth Connected Component + Planar

$$|V_1| + |R_1| = |E_1| + 2$$
$$|V_2| + |R_2| = |E_2| + 2$$
$$|V_3| + |R_3| = |E_3| + 2$$
$$\vdots$$
$$|V_k| + |R_k| = |E_k| + 2$$

$$\Sigma \Rightarrow (|V_1| + |V_2| + \cdots + |V_k|) + (|R_1| + |R_2| + \cdots + |R_k|) = (|E_1| + |E_2| + \cdots + |E_k|) + 2k$$

$|V_1| + |V_2| + \cdots + |V_k| \rightarrow |V|$

$|R_1| + |R_2| + \cdots + |R_k| \rightarrow |R| + (k-1)$

$|E_1| + |E_2| + \cdots + |E_k| \rightarrow |E|$

$$|V| + (|R| + k - 1) = |E| + 2k$$

$$\boxed{|V| + |R| = |E| + k + 1}$$

Euler's equation for disconnected Planar graph

{may or may not be simple}

Let 'G' is a planar graph with 'K' Connected Components.
{May not be simple}

then following Equation will hold true

$$|V| + |R| = |E| + K + 1$$

it is called Euler's Equation for disconnected planar graph

Where,

$|V|$ = No. of Vertices in G

$|E|$ = No. of edges in G

$|R|$ = No. of regions in G = (Total no. of interior regions + '1' Exterior region)

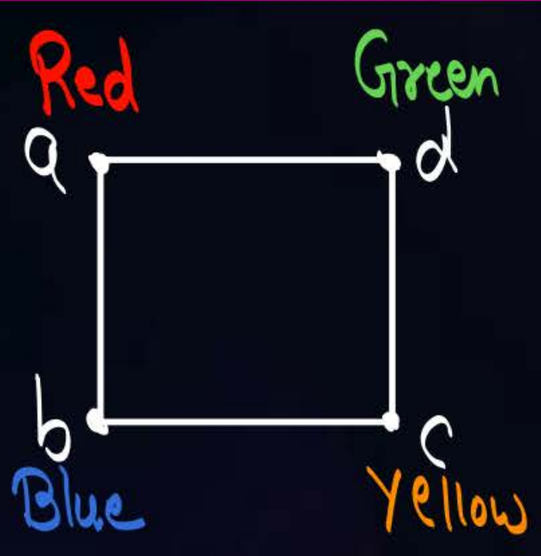
K = No. of Connected Components in G

Coloring

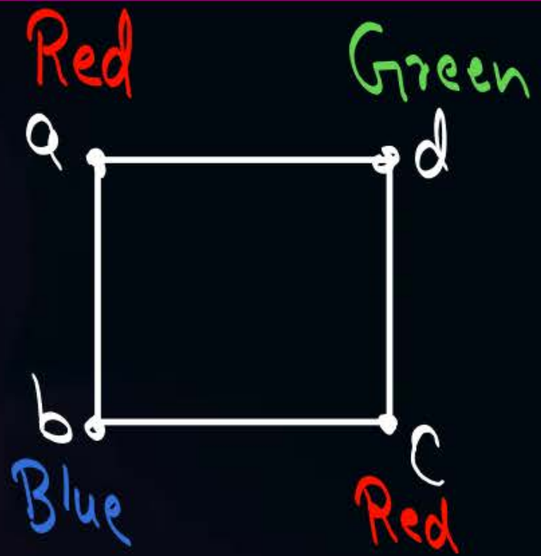
Topic : Vertex coloring

- An assignment of colors to the vertices of graph G, such that no two adjacent vertices of the graph have the same color is called vertex coloring of graph G.

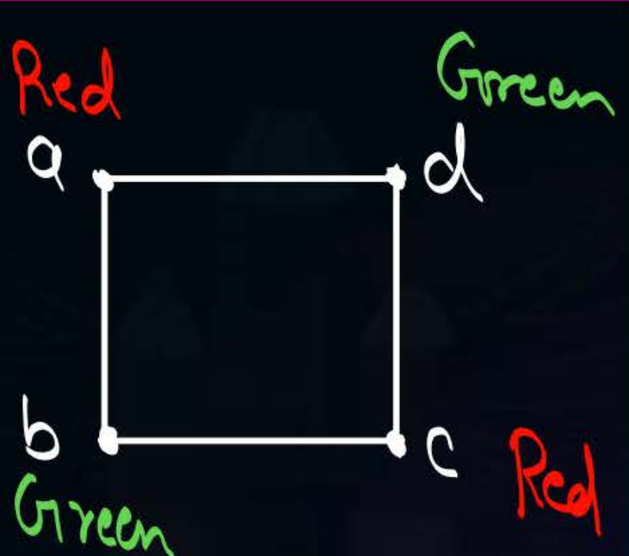
All of them are valid Vertex Coloring



Vertex Coloring using '4' Colors



Vertex Coloring using '3' Colors



Vertex Coloring using '2' Colors

→ If we have a graph with 'n' vertices, then we can always Vertex Color that graph using 'n' Colors {i.e. one color per vertex}

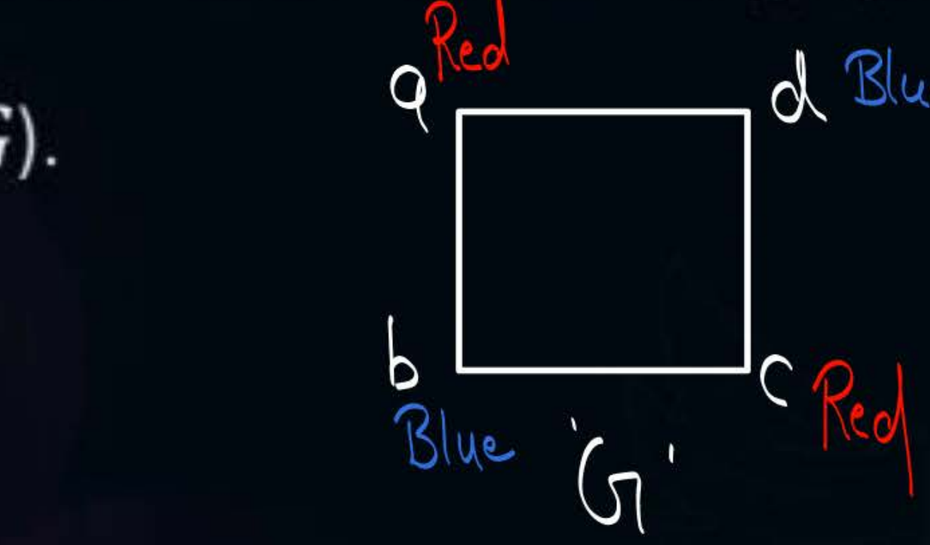
"But we are interested in the minimum number of Colors required for the Vertex Coloring of the given graph"



Topic : Chromatic number

Minimum number of colors needed for the vertex coloring of graph G is called chromatic number of graph G,

It is denoted by $\chi(G)$.



We can not vetex Color the graph G using less than '2' Colors

$\therefore \chi(G) = 2$



Topic : K-colorable vs K-chromatic

K-Colorable

- If there exist a vertex coloring of graph 'G' that uses 'K' colors, then graph G is called 'K' Colorable

If graph G is K-Colorable, then $\chi(G) \leq K$

K-Chromatic

If chromatic number of graph 'G' is 'K', then graph 'G' is said to be K-chromatic

If graph G is K-chromatic, then $\chi(G) = K$

Note:- ★

① If graph G is K-Colorable, then graph G is (K+1)-Colorable as well,

Provided $(K+1) \leq$ No. of vertices in graph G.

② If graph G is K-Colorable, then graph G may or may not be (K-1)-Colorable

③ If graph G is K-chromatic, then it is neither (K+1)-chromatic nor (K-1)-chromatic

Note:- ★

④ If graph G is k-chromatic then graph G is k-Colorable as well, but

if graph is k-Colorable then it need not be k-Chromatic



Topic : Four-Color Theorem

Every planar graph G is four colorable.

For any planar graph G, $\chi(G) \leq 4$

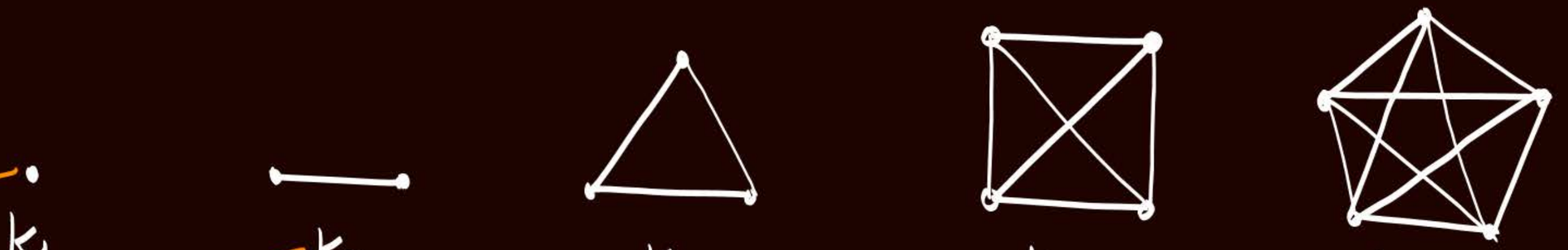
Note :→

① Chromatic number of graph G Can be '1' if and only if graph 'G' is a NULL graph

i.e. $(\chi(G)=1) \iff$ 'G' is NULL graph

② If graph G is not a NULL graph then $\chi(G) \geq 2$

Q:- Find the Chromatic number of Complete graph 'K_n'



K_1 K_2 K_3 K_4 K_5 . . .

$\chi(K_1) = 1$ $\chi(K_2) = 2$ $\chi(K_3) = 3$ $\chi(K_4) = 4$ $\chi(K_5) = 5$

For a Complete graph K_n, $\chi(K_n) = n$

Max degree

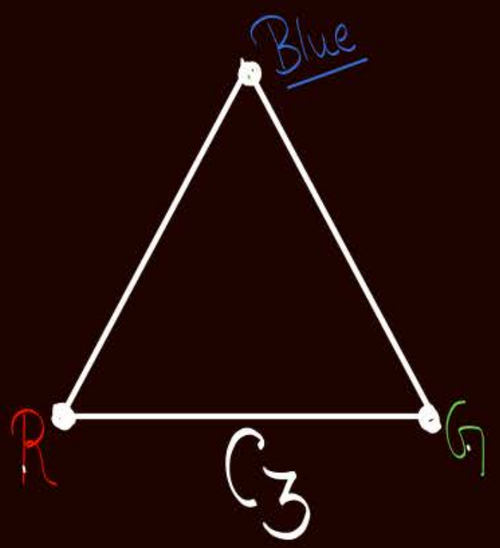
$\Delta(K_1) = 0$ $\Delta(K_2) = 1$ $\Delta(K_3) = 2$ $\Delta(K_4) = 3$ $\Delta(K_5) = 4$

$\chi(K_1) = 1 + \Delta(K_1)$ $\chi(K_2) = 1 + \Delta(K_2)$ $\chi(K_3) = 1 + \Delta(K_3)$ $\chi(K_4) = 1 + \Delta(K_4)$ $\chi(K_5) = 1 + \Delta(K_5)$

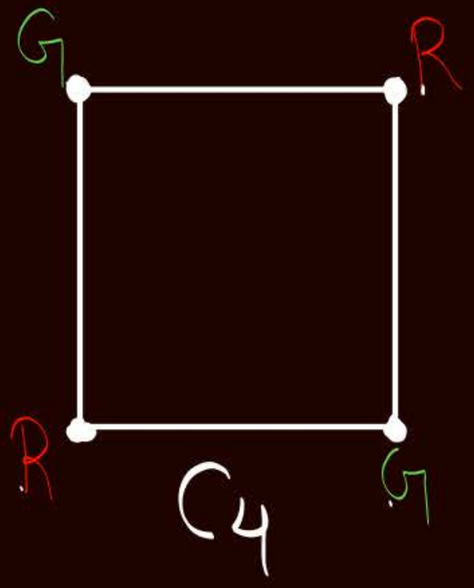
For any graph G,

$$\chi(G) \leq 1 + \Delta(G)$$

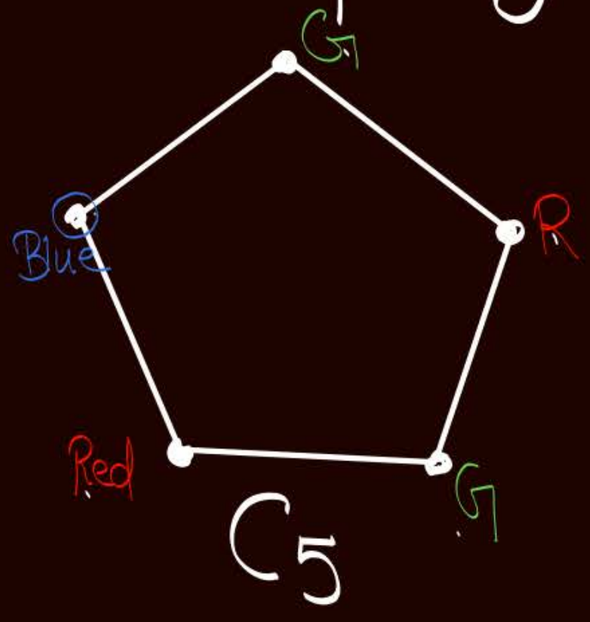
Q:- Find the Chromatic number of Cycle graph 'Cn'



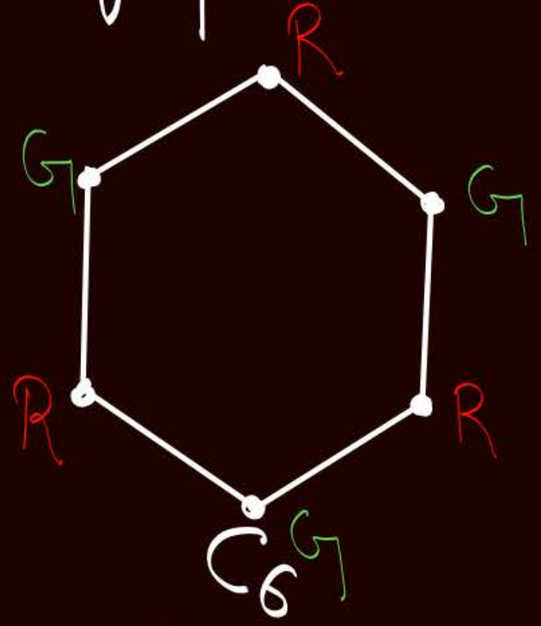
$\chi(C_3) = 3$



$\chi(C_4) = 2$



$\chi(C_5) = 3$

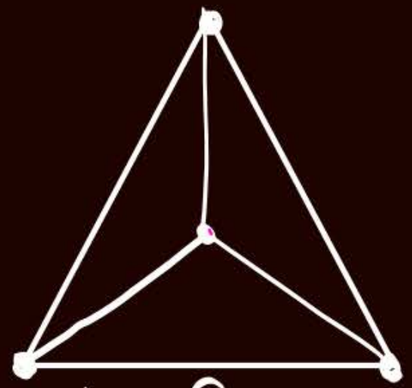


$\chi(C_6) = 2$

$$\chi(C_n) = \begin{cases} 2, & \text{if } n = \text{Even} \\ 3, & \text{if } n = \text{odd} \end{cases}$$

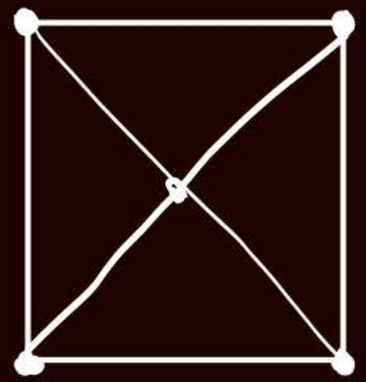
= it is also $(1 + \Delta(G))$

Q:- Find the Chromatic number of Wheel graph W_n



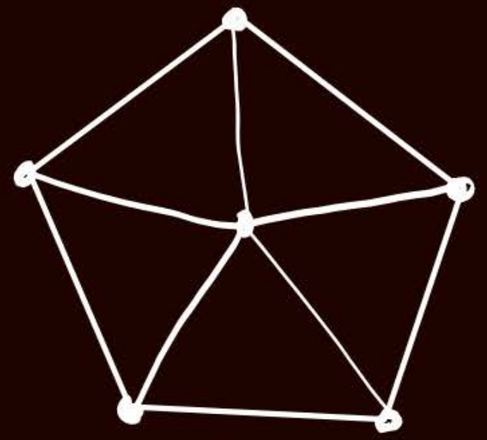
$W_4 = C_3 + 1 \text{ hub}$

$\chi(W_4) = 3 + 1 = 4$



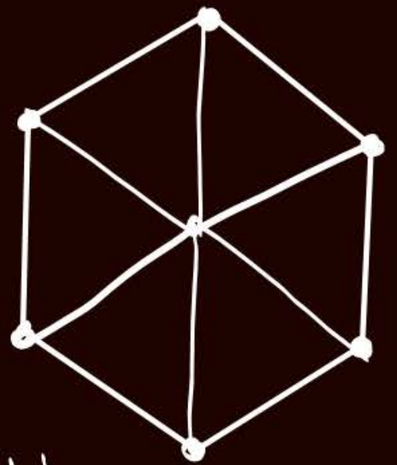
$W_5 = C_4 + 1 \text{ hub}$

$\chi(W_5) = 2 + 1 = 3$



$W_6 = C_5 + 1 \text{ hub}$

$\chi(W_6) = 3 + 1 = 4$



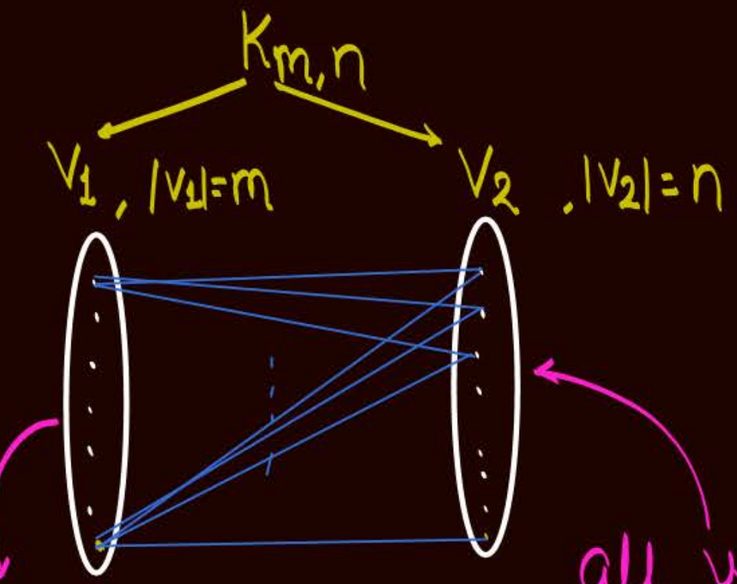
$W_7 = C_6 + 1 \text{ hub}$

$\chi(W_7) = 2 + 1 = 3$

Hub vertex will always require a different color.

$$\chi(W_n) = \begin{cases} 4, & \text{if } n = \text{even} \\ 3, & \text{if } n = \text{odd} \end{cases}$$

Q:- Find the Chromatic number of Complete bipartite graph $K_{m,n}$



i.e. One Color per Subset of Partition

$\therefore \chi(K_{m,n}) = 2$

Note:

① Complete bipartite graph can never be a NULL graph because no subset of partition can be Empty. and Each vertex of 1st non-Empty subset will Connect with each vertex of 2nd non-Empty subset. Hence at least one Edge will be present in the graph

But, a bipartite graph may be a NULL graph

eg.



it is a bipartite graph, And no Edge ∴ NULL graph

① For a Complete bipartite graph Chromatic no. is always = 2

② But for a bipartite graph Chromatic no. is 1 or 2

1 → when NULL graph

2 → otherwise

Note:- ① Chromatic no. of a bipartite graph is 1 or 2

i.e. χ (Bipartite graph) ≤ 2

② Bipartite graph are also known as bi-Colorable graph {i.e. 2-Colorable}, but not 2-chromatic

③ If Chromatic number of a graph 'G' is '2' then graph 'G' is a bipartite graph.

i.e. if $\chi(G) = 2$, then G is bipartite



2 mins Summary

Topic Euler's equation for disconnected planar graph

Topic Vertex coloring

Topic Chromatic number

Topic Edge coloring

THANK - YOU